Do Now · retrieval

Input and output devices

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why is a GPU suitable for applying the same filter to many independent pixels?

Q2

What role does a processor's control unit have?

Q3

What data type should represent the state of a binary switch?

Q4

ASCII encodes A as decimal 65 and B as 66. A string 'ABBA' uses four 8-bit stored code units in this simplified example. How many bits and bytes, and why is this not proof all Unicode characters fit in one byte?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Select suitable input — A temperature sensor measures a physical quantity; an analogue-to-digital converter is needed if the controller uses digital samples.

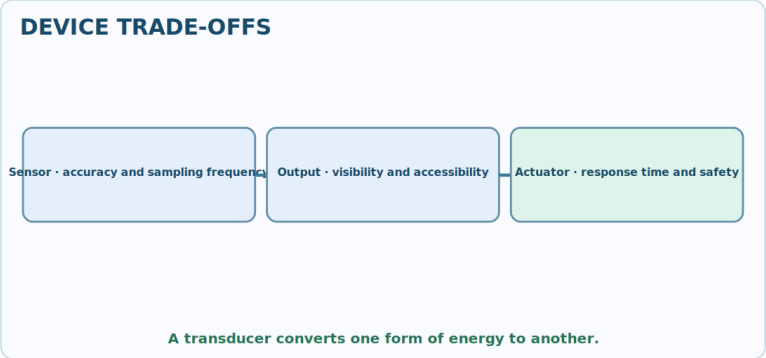
Core idea 2

Process and control — Software compares the sampled temperature with a target to decide whether to open the vent.

Core idea 3

Choose appropriate outputs — An actuator makes the physical change; a display can show readings. Design must address accessibility and suitable latency.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

sensor — Input transducer measuring a physical quantity.

ADC — Analogue-to-Digital Converter.

actuator — Output component that causes physical action.

latency — Delay between input and system response.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

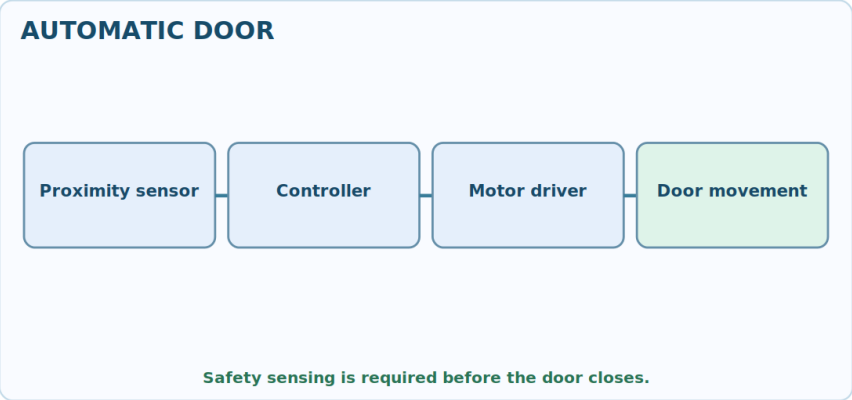
Choose input and output for an automatic door that opens when someone approaches.

Follow the worked method

1. Predict first — A proximity or motion sensor detects someone approaching and passes a signal to the controller.

2. Explain and check — A motor and driver create movement; safety sensing reduces the risk of closing onto a person.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
A push button is a clear intentional input and the controller processes it under safe traffic timing rules.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Name a suitable input and output device for a weather station, and explain the role of each.

Question 2

A greenhouse opens a vent when temperature exceeds a limit. Name the sensor, processing decision and actuator, and explain any analogue conversion.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate device choices for a pedestrian crossing serving users with hearing or visual impairment.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A motor that physically opens a door is an input device.

Use key vocabulary from memory

1. Explain sensor in your own words.

2. Explain ADC in your own words.

3. Explain actuator in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.